

2024, Mar. 2024 Release LOC:Acute Pediatric

Brief Resolved Unexplained Event (BRUE)

Utilization Benchmarks

Length of Stay: 1.3 d

Length of Stay Type: InterQual

Condition/Procedure: BRIEF RESOLVED UNEXPLAINED EVENT

% Paid as Observation: 71%

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Overview

Select Day

Episode Day 1

Episode Day 2 ⁽¹²⁾

Episode Day 3

Discharge Screens ⁽¹⁸⁾

Notes

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Overview

Instruction: This subset is for infants less than 1-year old with a brief resolved unexplained event (BRUE) who need further evaluation and diagnostic workup.

Patients who are found to have actual or suspected gastroesophageal reflux disease, seizures, cyanosis with suspected cardiac etiology, or an arrhythmia can be reviewed in the General Medical subset. Patients who have evidence of abuse and neglect resulting in traumatic injury can be reviewed in the General Trauma subset. Patients who have epilepsy can be reviewed in the Epilepsy subset. Patients who are found to have an infection can be reviewed in the appropriate infection specific subset.

Introduction: A BRUE, formally known as ALTE (apparent life-threatening event), is diagnosed when an infant younger than 1-year old experiences a sudden and unexplained event of at least one of the following:

- Cyanosis or pallor
- Absent, decreased, or irregular breathing
- Hypertonia or hypotonia
- Altered level of responsiveness

A BRUE is an unexplained event of unknown etiology in infants less than 1-year old. If the infant has additional symptoms (i.e., abnormal vital signs, cough, respiratory difficulties, fever), it is not a BRUE and an alternative etiology should be sought (Tieder et al., Pediatrics 2016, 137:).

Evaluation and Treatment: A thorough history and physical exam helps guide the management of the infant. The information obtained may provide clues to the etiology and severity of the BRUE, helping to focus the subsequent diagnostic evaluation. For patients who have experienced a BRUE, the risk of reoccurrence and presence of an underlying etiology should be determined. A thorough medical, family, social, and environmental history should be obtained (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most

appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:).

(Excludes PO medications unless noted)

Select Day, One:**● Episode Day 1, One:**

(Symptom or finding within 24h)

● OBSERVATION, All: ⁽¹⁾**● BRUE, actual or suspected and, ≥ One:** ⁽²⁾

- Absent, decreased, or irregular breathing ⁽³⁾
- Altered level of responsiveness ⁽⁴⁾
- Cyanosis or pallor ⁽⁵⁾
- Hypotonia or hypertonia

● Risk factor, ≥ One: ⁽⁶⁾**● Abuse or neglect, actual or suspected and, ≥ One:**

- Investigation and family assessment or therapy ⁽⁷⁾

● Placement in protective custody or foster care indicated and, ≥ One:

- Arrange placement
- Arrange follow-up care
- Education of custodial caregiver
- Age ≤ 60d
- Born < 32 wks gestation and postconceptional age < 45 wks ^(8, 9)
- CPR required by trained medical provider
- BRUE duration > 1 min
- Event occurring in clusters
- Family history of sudden unexplained death
- History of > 1 BRUE ⁽²⁾

● Intervention, All:

- Continuous cardiac monitoring (excludes Holter)
- Continuous oximetry ⁽¹⁰⁾
- Diagnostic evaluation and monitoring ⁽¹¹⁾

● Episode Day 2, One: ⁽¹²⁾**● OBSERVATION, One:** ⁽¹⁾**● Responder, discharge expected today if clinically stable last 12h, All:** ⁽¹³⁾

- Diagnostic evaluation and monitoring complete
- No recurrent BRUE

● Parent or caregiver demonstrates ability to provide care, Both:

- Discharge teaching complete ⁽¹⁴⁾
- Home environment safe or alternative living arranged ^(15, 16)
- Vital signs stable or within acceptable limits ⁽¹⁷⁾

● Partial responder, not clinically stable for discharge and requires continued stay, Both:**● Abuse or neglect, actual or suspected and, ≥ One:**

- Investigation and family assessment or therapy ⁽⁷⁾

● Placement in protective custody or foster care indicated and, ≥ One:

- Arrange placement
- Arrange follow-up care
- Education of custodial caregiver

● Intervention, Both:

- Continuous cardiac monitoring (excludes Holter)
- Continuous oximetry

● Episode Day 3, One:**● OBSERVATION, One:** ⁽¹⁾**● Responder, discharge expected today if clinically stable last 12h, All:** ⁽¹³⁾

- Diagnostic evaluation and monitoring complete
- No recurrent BRUE

● Parent or caregiver demonstrates ability to provide care, Both:

- Discharge teaching complete ⁽¹⁴⁾
- Home environment safe or alternative living arranged ^(15, 16)
- Vital signs stable or within acceptable limits ⁽¹⁷⁾
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One**:
 - For a condition other than BRUE use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review

Discharge, Level of Care, ≥ One: ⁽¹⁸⁾● **HOME, Both:**● Level of care appropriateness, **Both**:

- Home environment safe and accessible ⁽¹⁹⁾
- Patient or caregiver demonstrates ability to manage care ⁽²⁰⁾

● BRUE discharge planning, **All**:

- Provide comprehensive written discharge and teaching instructions ⁽²⁰⁾
- Education on the importance of CPR training and early recognition of signs of BRUE ⁽¹⁴⁾
- Medication reconciliation ⁽²¹⁾
- Follow-up care planned ^(22, 23)
- Identify and address transportation needs

● **HOME CARE, Both:**● Level of care appropriateness, **All**:

- Home environment safe and accessible ⁽¹⁹⁾
- Patient or caregiver willing and able to learn care
- Outpatient management contraindicated or unavailable ^(24, 25)

● Skilled services, ≥ **One**: ⁽²⁶⁾

- Skilled nursing ⁽²⁷⁾
- Skilled therapy, PT, OT, or ST
- Behavioral health ⁽²⁸⁾

● BRUE discharge planning, **All**:

- Provide comprehensive written discharge and teaching instructions ⁽²⁰⁾
- Education on the importance of CPR training and early recognition of signs of BRUE ⁽¹⁴⁾
- Medication reconciliation ⁽²¹⁾
- Follow-up care planned ⁽²²⁾
- Identify and address transportation needs

● **BEHAVIORAL HEALTH, Both:**● Level of care appropriateness, **One**:

- Support systems available ⁽²⁹⁾

● Skilled treatment, ≥ **One**:

- Clinical assessment ⁽³⁰⁾
- Individual, group, or family counseling
- Psychiatric, substance, or medication evaluation ⁽³¹⁾

● **SKILLED MEDICAL OR THERAPY, Both:**● Level of care appropriateness, **All**:

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One**:

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽²⁷⁾

● Therapy, **All**:

- Goal directed therapy with rehabilitation potential ^(32, 33)
- Functional impairments requiring at least supervision ⁽³⁴⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All**:

- Medication reconciliation ⁽²¹⁾

- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE MEDICAL OR THERAPY, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 2x/wk$
- Treatment precluded in a lower level

● Services, $\geq$ One:

- Medical and skilled nursing interventions or assessment 4h/24h ⁽²⁷⁾

● Therapy, All:

- Goal directed therapy with rehabilitation potential ^(32, 33)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁴⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 1 therapy discipline

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽²¹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● SUBACUTE REHAB, Both:**● Level of care appropriateness, All:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
- Treatment precluded in a lower level

● Services, All:

- Goal directed therapy with rehabilitation potential ^(32, 33)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁴⁾
- Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽²¹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● ACUTE REHAB, Both:**● Level of care appropriateness, Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$

● Services, All:

- Goal directed therapy with rehabilitation potential ^(32, 33)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽³⁴⁾
- Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽³⁵⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽²¹⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● LONG TERM ACUTE CARE, Both:**● Level of care appropriateness, Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ One:

- Primary condition or illness and treatment of active comorbid condition(s) ⁽³⁶⁾
- Ventilator dependent $\geq 6h/d$ and weaning planned ^(37, 38)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽³⁹⁾

● Complete prior to facility transfer, All:

- Medication reconciliation ⁽²¹⁾
- Obtain and complete forms for facility

- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Notes:**1:**

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

2:

A brief resolved unexplained event (BRUE) is defined as an event occurring when an infant experiences a brief and now-resolved episode of one or more of the following (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:):

- Absent, decreased, or irregular breathing
- Altered level of responsiveness
- Cyanosis or pallor
- Marked change in tone (hyper- or hypotonia)

3:

Instruction: Absent, decreased, or irregular breathing includes patients who experienced a prolonged episode of unexplained apnea.

4:

Altered level of responsiveness specific to a brief resolved unexplained event (BRUE) includes: loss of consciousness, mental status change, lethargy, and/or somnolence. An altered level of responsiveness may indicate a serious underlying condition (e.g., cardiac, respiratory, metabolic, neurologic event) (Tieder et al., Pediatrics 2016, 137:).

5:

Central cyanosis describes a blue or purple coloration of the face, gums, or trunk caused by low circulation of oxygenated hemoglobin. Central pallor describes a pale coloration of the face or trunk (Tieder et al., Pediatrics 2016, 137:).

6:

Guidelines suggest that the following are risk factors for patients presenting with a brief resolved unexplained event (BRUE) (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:):

- ≤ 60 days old
- Child abuse, actual or suspected
- Born < 32 weeks gestation and postconceptional age < 45 weeks
- CPR required by trained medical provider
- Duration of BRUE > 1 minute
- Event occurring in clusters
- Family history of sudden unexplained death
- History of > 1 BRUE

7:

During the family assessment, a thorough clinical assessment based on the clinician's observations of how the family functions should be gathered by a social worker. At this time, the patient's history and history at home or at a day-care center should also be obtained. This initial family therapy session and/or assessment should ideally be conducted to facilitate a safe discharge.

8:

Postconceptional age is expressed in weeks and is defined as the gestational age plus the number of weeks since

birth (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:).

9:

Guidelines suggest that prematurity of < 32 weeks gestational age and < 45 weeks postconceptional age would put an infant presenting with a brief resolved unexplained event at higher-risk in need of further monitoring and testing (Merritt et al., Pediatrics 2019, 144: epub).

10:

Infants with Brief Resolved Unexplained Event (BRUE) need not have laboratory values, imaging studies, and other diagnostic procedures, however, a period of cardiopulmonary, oximetry monitoring and serial examinations may be warranted to ensure stability of vital signs and no recurrent BRUE (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:).

11:

Patients with a suspected brief resolved unexplained event will need a thorough physical assessment and evaluation of systems to rule out possible etiologies. Additional evaluation based on the characteristics of the event may include (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:):

- Assessment for episodic hypoglycemia or acidosis
- Cardiology consultation
- Concern for seizures
- Feeding studies
- Gastroenterology consultation

12:

Instruction: If a patient is not suspected of being abused or neglected but has another suspected etiology for the brief resolved unexplained event (BRUE), they should be reviewed in the appropriate condition specific subset. Patients who are found to have actual or suspected gastroesophageal reflux disease, cyanosis with suspected cardiac etiology, seizures or an arrhythmia can be reviewed in the General Medical subset. Patients who have evidence of abuse and neglect resulting in traumatic injury can be reviewed in the General Trauma subset. Patients who have epilepsy can be reviewed in the Epilepsy subset. Patients who are found to have an infection can be reviewed in the appropriate infection specific subset.

Patients who are found to have an actual BRUE, should continue to be reviewed within the BRUE subset.

13:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

14:

Upon discharge caregivers should be offered resources for cardiopulmonary resuscitation training and educated on early recognition of the signs and symptoms of brief resolved unexplained events (Merritt et al., Pediatrics 2019, 144: epub; Tieder et al., Pediatrics 2016, 137:).

15:

A safe home environment includes running water, electricity, phone, temperature control (hot or cold), and access to transportation.

16:

Any suspected abuse or neglect should be ruled out prior to discharge to the home setting or alternative living arrangements should be in place.

17:

Vital sign stability refers to the blood pressure (systolic and diastolic), heart rate, and respiratory rate being within

the normal ranges for the pediatric patient's age. When criteria states "within acceptable limits" it refers to either the patient's normal baseline, newly established baseline, or to parameters which the medical practitioner determines are acceptable.

18:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

19:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

20:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

21:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

22:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

23:

Outpatient follow-up is recommended to identify infants with medical concerns warranting further treatment (Tieder et al., Pediatrics 2016, 137:).

24:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

25:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

26:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

27:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

28:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

29:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

30:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

31:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

32:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work)

environment)

- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

33:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

34:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

35:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

36:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

37:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

38:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

39:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784).

Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.